Two-way tables

These are a great way to display all the pairs of outcomes for two events, actions or questions.

For example: A group of students where asked if they had ever gone surfing or kayaking before.

There are four different possible responses (outcomes) each student can give.

Yes to surfing and yes to kayaking
Yes to surfing and no to kayaking
No to surfing and yes to kayaking
No to surfing and no to kayaking

We can represent these outcomes in a two-way table.

		Kayaking	
		Yes	No
Surfing	Yes	← Yes ↑ Yes Students here have done both activities	← Yes ↑ No Students here only have been surfing
	No	← No ↑ Yes Students here have only been kayaking	← No ↑ No Students here have done neither activity

Each cell showing the outcome pairings is filled with the frequency of that outcome like this:

		Kayaking	
		Yes	No
Surfing	Yes	8	6
	No	4	12

The total number of students surveyed is the sum of all the values in the two-way table $(= 30)$.

Numbered red and black cards were selected at random and the outcomes recorded

		Number	
		Odd	Even
Color	Black	11	16
	Red	14	9

(i) How many cards were selected?

Number of cards selected $= 11 + 16 + 14 + 9$

$= 50$

(ii) How many red cards were selected with an odd number on them?

$n(Red, Odd) = 14$

(iii) How many black cards were selected?

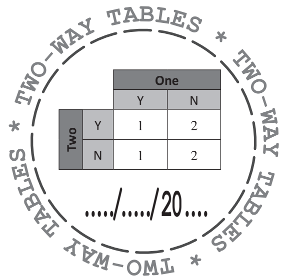
$n(Black) = 11 + 16$

$= 27$

(iv) Which outcome had the highest frequency?

A black card with an even number;
$n(Black, Even) = 16$

Two-way tables



1. Complete the two-way tables for each of these:

   a. People were asked if they preferred their apple pie hot/cold and with/without ice cream.

		Ice Cream	
Apple pie			

   b. A decision spinner (Yes/No) spun either clockwise or counter-clockwise.

		Direction	

   c. A new local sports team asked its players for their preferred choice of team colours between black/white shorts and yellow, red or orange shirts.

2. Answer the questions for the two-way table below showing the results of a musical survey.

		Play guitar	
		Yes	No
Play flute	Yes	5	12
	No	14	23

   a. How many people were surveyed?

   b. How many people surveyed play both instruments?

   c. How many people surveyed play the flute?

   d. How many people surveyed can play one instrument only?

   e. $n(Play\ guitar, Do\ not\ play\ flute) =$

   f. Another 5 people who do not play either instrument are surveyed. What change needs to be made to the table?

Two-way tables

3 Aruma School has 200 students who travel from four surrounding towns. The towns where all the boys and girls in this school live was recorded in a two-way table.

	Girls	Boys
Neeuk Creek	18	21
Nooroon Plains	37	26
Dilkara	15	
Alba	42	33

a How many boys travel from Dilkara to attend the school?

b How many girls attend Aruma School?

c A large family with two sons and four daughters who attend Aruma School move from Nooroon Plains to Neeuk Creek. Complete the two-way table with the new values below.

	Girls	Boys
Neeuk Creek		
Nooroon Plains		

4 A group of people took part in a blind tasting test as part of a market research.
Each person had to say whether brand A or brand B tasted better. Then they needed to decide if the brands were better on their own, or with ingredient X or ingredient Y added.
Use the information collected to fill in the two-way table.

- 10 people preferred brand B without any added ingredients.
- Four times as many people preferred brand B with ingredient Y added, than on its own.
- Forty seven people preferred the product after ingredient Y was added.
- Thirty four people preferred the brands unchanged.
- An equal amount of people liked Brand A with nothing added and with ingredient X added.
- 120 people participated.

	Nothing Added	Ingredient	Ingredient
Brand:			
Brand:			

Two-way table probabilities

The fraction of observed results in any experiment/collection of trials is called the relative frequency.

Number of times it happens

$$\text{Relative frequency} = \frac{\text{The frequency of the outcome being observed}}{\text{The number of trials completed}}$$

The values along with the row and column totals make relative frequency calculations easy.

Complete the two-way table and use it to answer the given questions below

After tossing a coin and rolling a four-sided die together 100 times, the tally of each outcome pairing was recorded.

$(H, 1) =$ 𝍸 𝍸 ||| $(H, 2) =$ 𝍸 ||| $(H, 3) =$ 𝍸 𝍸 || $(H, 4) =$ 𝍸 𝍸 ||||

$(T, 1) =$ 𝍸 𝍸 𝍸 $(T, 2) =$ 𝍸 𝍸 |||| $(T, 3) =$ 𝍸 ||| $(T, 4) =$ 𝍸 𝍸 𝍸 |

(i) Record the observed results into a two-way table.

		4 sided die					
		1	2	3	4	**Total**	
Coin	Hand (H)	13	8	12	14	47	← Total heads
	Tail (T)	15	14	8	16	53	← Total tails
	Total	28	22	20	30	100	← Sum of the column/row totals should both equal the total number of trials.
		Total 1s	Total 2s	Total 3s	Total 4s		

(ii) Calculate the relative frequency and theoretical probability for the outcome $(T, 4)$.

Relative Frequency (or Experimental Probability)

$n(T, 4) = 16, \quad n(Trials) = 100$

Relative frequency of $(T, 4) = \dfrac{n(T, 4)}{n(Trials)} = \dfrac{16}{100} = \dfrac{4}{25}$

Theoretical Probability

$n(T, 4) = 1, \quad n(S) = 8$

$P(T, 4) = \dfrac{n(T, 4)}{n(S)} = \dfrac{1}{8}$

The more trials completed, the closer we expect the relative frequency to match the theoretical probability.

(iii) Calculate the relative frequency for flipping a Head when the number 3 was rolled.

$$n(H, 3) = 12, \quad n(Trials\ in\ which\ a\ 3\ was\ rolled) = 20$$

Relative frequency of heads when a 3 is rolled $= \dfrac{n(H, 3)}{n(Trials\ in\ which\ a\ 3\ was\ rolled)}$

$= \dfrac{12}{20}$

$= \dfrac{3}{5}$ Simplified form

$= 60\%$ Percentage form

Two-way table probabilities

1. Complete these two-way tables using the collected data given.

a. The position of two light switches observed 20 times.

($Switch$ 1, $Switch$ 2)

(On, On): ||||
(On, Off): 𝍸
(Off, On): |||
(Off, Off): 𝍸 |||

		Switch 2		
		On	Off	Total
Switch 1	On			
	Off			
	Total			

b. The results of flipping two coins 50 times.

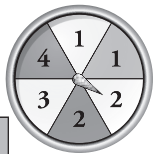
($Coin$ 1, $Coin$ 2)

(H, H): 𝍸 |||
(H, T): 𝍸 𝍸 ||||
(T, T): 𝍸 𝍸 𝍸 ||
(T, H): 𝍸 𝍸 |

		Coin 2		
		Head (H)	Tail (T)	Total
Coin 1	Head (H)			
	Tail (T)			
	Total			

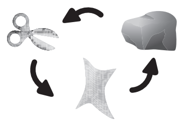
2. a. Fill in the missing values for these two-way tables:

(i) Shades and numbers spinner.



	White	Grey	Total
1	18	15	
2		11	30
3	13		22
Total		35	

(ii) Scissors, Paper, Rock games.



		Player 2			
		Scissors (S)	Paper (P)	Rock (R)	Total
Player 1	Scissors (S)	8	9	5	
	Paper (P)	11		7	23
	Rock (R)	4	6		15
	Total		20		60

b. Use the completed tables to answer these:

(i) How many spins were observed on the shades and numbers spinner?

(ii) How many games of Scissors, Paper, Rock were recorded?

(iii) How many times did the spinner stop on a grey sector?

(iv) How many times did player 1 say 'paper'?

(v) How many times did a game have scissors and paper (by either player) as the result?

(vi) If rock beats scissors, which player won the most games when this outcome occurred?

(vii) How many times did Player 1 and Player 2 make the same symbol?

Two-way table probabilities

3 This two-way table shows the results of a random survey of students in a school who were asked a yes/no question followed by a multiple choice question.

		Q2				
		A	B	C	D	Total
Q1	Yes (Y)	4	7	1	0	12
	No (N)	8	17	10	3	38
	Total	12	24	11	3	50

a) How many students were surveyed in this school?

b) How many students surveyed selected 'C' for Question 2?

c) What was the most common outcome for the two questions asked in this survey?

d) What outcome did not occur for the two questions asked in this survey?

e) What is the frequency for the outcome 'Yes, A'?

f) What is the relative frequency for the outcome 'Yes, A'?

g) What is the relative frequency for an answer of 'No' to Q1 as a percentage?

4 Numbers 1 through to 20 were printed on two packs of twenty cards. One pack printed using red ink (R) and the other printed using green ink (G). The two packs were then shuffled together.

a) Twenty four cards were randomly selected (with replacement) and the outcomes recorded. Complete the two-way table given using the recorded outcomes below.

$(G, 15)$	$(G, 1)$	$(G, 11)$	$(R, 6)$
$(R, 5)$	$(R, 18)$	$(R, 3)$	$(G, 17)$
$(R, 12)$	$(G, 8)$	$(R, 19)$	$(R, 2)$
$(G, 7)$	$(R, 1)$	$(G, 3)$	$(G, 6)$
$(G, 10)$	$(R, 3)$	$(G, 5)$	$(G, 6)$
$(R, 14)$	$(R, 16)$	$(G, 13)$	$(R, 5)$

		Colour		
		Red	Green	Total
Number	≤ 10			
	>10			
	Total			

b) Calculate the expected (theoretical) probability of selecting a green card with a number ≤ 10 and its relative frequency following the random selections.

c) If the number of random selections is increased greatly, what do you expect will happen to the theoretical probability and relative frequency values?

Two-way table probabilities

5 Children who successfully tossed a ring around a particular bottle at a local fair where then given the chance to pick a prize randomly from the lucky dip box.
The lucky dip box contained two different prizes: A mini mp3 player or a packet of bubble gum.
After one day at the fair, the results in the two-way table below were observed.

		Lucky Dip		
		Mp3	Gum	**Total**
Ring toss successful	Yes (Y)	2	48	50
	No (N)	0	0	0
	Total	2	48	50

a Explain why the values for ‘No’ are all zero.

b Write the relative frequency of each lucky dip prize in percentage form.

c Explain why you think there is a large difference between the two values in part b.

d After another 30 successful ring tosses, all the prizes drawn from the lucky dip were bubble gum. What is the new relative frequency for each lucky dip prize following the extra 25 wins?

6 Earn an Awesome passport stamp for this one.
Use the information given here to complete the two-way table below

- Relative frequency of the outcome $(Even, Up) = \dfrac{32}{115}$
- 64 Odd numbers were observed
- Relative frequency of the direction being ‘Up’ when an odd number was observed $= 25\%$

		Direction		
		Up	Down	**Total**
Number	Even			
	Odd			
	Total